

Netflix Exploratory Data Analysis (EDA)

Cell 1: Markdown

Netflix Exploratory Data Analysis (EDA)

Problem Statement

Netflix is one of the world's leading streaming platforms, offering thousands of movies and TV shows across different countries, genres, and age ratings. As the content library continues to grow, understanding patterns in the dataset becomes essential for making informed business decisions related to content acquisition, audience preferences, and market expansion.

The objective of this project is to perform an in-depth Exploratory Data Analysis (EDA) on the Netflix Titles dataset using Python. The analysis aims to clean and preprocess the data, identify trends and patterns, uncover meaningful insights, and visualize key information about Netflix's content catalog.

This project focuses on answering important business questions such as:

- What is the distribution of Movies and TV Shows on Netflix?
- Which countries contribute the most content?
- Which genres dominate the platform?
- How has Netflix's content library grown over the years?
- What are the most common content ratings?
- Which directors and actors appear most frequently?
- What are the trends in content additions over time?
- What insights can help understand Netflix's content strategy?

By leveraging Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, this project demonstrates the complete data analysis workflow, including data cleaning, transformation, visualization, and business insight generation.

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Project Objectives

- Import and explore the Netflix dataset.
- Perform comprehensive data cleaning and preprocessing.
- Handle missing values and duplicate records.
- Convert and transform data into suitable formats.
- Create new features for deeper analysis.
- Conduct univariate, bivariate, and multivariate analysis.
- Visualize trends using professional charts.
- Answer business-oriented analytical questions.
- Generate actionable insights based on the findings.
- Build a well-documented, portfolio-ready EDA project using Python.

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Dataset Description

The dataset contains information about movies and TV shows available on Netflix, including:

- Show ID
- Content Type (Movie / TV Show)
- Title

- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre (Listed In)
- Description

These attributes enable comprehensive analysis of Netflix's content catalog from multiple business perspectives.

Cell 4: Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

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Import Data

Cell 6: Code

```
df = pd.read_csv(r"C:\Users\Sumit Rathod\Downloads\netflix_titles.csv\netflix_titles.csv")
df
```

Cell 7: Markdown

Basic Exploration

Cell 8: Code

```
df.head()
```

Cell 9: Code

```
df.tail()
```

Cell 10: Code

```
df.sample(10)
```

Cell 11: Code

```
df.shape
```

Cell 12: Code

```
df.columns
```

Cell 13: Code

```
df.dtypes
```

Cell 14: Code

```
df.describe()
```

Cell 15: Code

```
(df.isnull().sum()/len(df))*100 # To check null values in percentage.
```

Cell 16: Markdown

- `director`,`cast`,`country`,`date\_added`,`rating`,`duration` this columns have null values.

Cell 17: Code

```
df.duplicated().sum()
```

Cell 18: Code

```
df.nunique()
```

Cell 19: Code

```
df['type'].value_counts()
```

Cell 20: Code

```
df['rating'].value_counts()
```

Cell 21: Code

```
df['country'].value_counts()
```

Cell 22: Markdown

Data Cleaning

Cell 23: Code

```
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')
```

Cell 24: Code

```
#Extracting.
```

Cell 25: Code

```
df['year'] = df['date_added'].dt.year
```

Cell 26: Code

```
df['month'] = df['date_added'].dt.month
```

Cell 27: Code

```
df['day'] = df['date_added'].dt.day
```

Cell 28: Code

```
#filling missing values in country.  
df['country'].fillna('unknown', inplace=True)
```

Cell 29: Code

```
df['director'].fillna('unknown', inplace=True)
```

Cell 30: Code

```
df.dropna(subset=['cast', 'date_added', 'rating', 'duration'], inplace=True)
```

Cell 31: Code

```
df.shape
```

Cell 32: Code

```
# After Cleaning the data .  
(df.isnull().sum()/len(df))*100
```

Cell 33: Markdown

Feature Engineering

Cell 34: Code

```
movies=df[df['type']=="Movie"]  
  
movies['duration']=movies['duration'].str.replace(" min","")  
movies['duration']=movies['duration'].astype(int)
```

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Univerarite

Cell 36: Code

```
sns.set_theme(style='darkgrid', palette='OrRd_r')
```

Cell 37: Code

```
plt.bar(df['type'].value_counts().index, df['type'].value_counts().values)  
plt.xlabel('Type')  
plt.ylabel('No of Movies/TV Shows')  
plt.show()  
plt.savefig('bar.jpeg')
```

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```
plt.figure(figsize=(15,7))  
plt.bar(df['rating'].value_counts().index, df['rating'].value_counts().values)  
plt.xlabel('Ratings')  
plt.ylabel('No of Ratings')  
plt.title('Rating Distribution', size=20)  
plt.show()
```

Cell 39: Code

```
top_10_country = df['country'].value_counts().head(10)
```

Cell 40: Code

```
plt.figure(figsize=(15,7))  
plt.bar(top_10_country.index, top_10_country.values)  
plt.title('Top 10 Countries producing Movies and TV Shows', size=20)  
plt.show()
```

Cell 41: Code

```
plt.figure(figsize=(15,7))
plt.bar(df['release_year'].value_counts().index, df['release_year'].value_counts().values)
plt.title('Release year Distribution', size=20)
plt.show()
```

Cell 42: Code

```
top= df['director'].value_counts().head(10)
```

Cell 43: Code

```
plt.figure(figsize=(20,7))
plt.bar(top.index, top.values)
plt.title('Top 10 Directors', size=20)
```

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Bivariate Analysis

Cell 45: Code

```
plt.figure(figsize=(15,7))
sns.countplot(x= df['type'], hue= df['rating'])
plt.title('Rating vs Type', size=20)
plt.show()
```

Cell 46: Code

```
movies = df[df['type'] == 'Movie'].copy()
```

Cell 47: Code

```
movies['duration'] = ( movies['duration'] .str.extract(r'(\d+)')[0].astype(float))
```

Cell 48: Code

```
avg_duration = (movies.groupby('release_year')['duration'].mean().reset_index())
```

Cell 49: Code

```
sns.lineplot(data=avg_duration, x='release_year', y='duration', marker='o')

plt.title('Average Movie Duration by Release Year', fontsize=15, fontweight='bold')
plt.xlabel('Release Year')
plt.ylabel('Average Duration (Minutes)')
plt.show()
```

Cell 50: Code

```
sns.pairplot(df)
```

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Business Questions

Cell 52: Code

```
# Total Movies?
df['type'].value_counts().head(1)
```

Cell 53: Code

```
#Total TV Shows?
df['type'].value_counts().tail(1)
```

Cell 54: Code

```
#Latest release year?
df['release_year'].sort_values(ascending=False).head(1)
```

Cell 55: Code

```
#Oldest release?
df['release_year'].sort_values(ascending=True).head(1)
```

Cell 56: Code

```
#Most common rating?
df['rating'].value_counts().sort_values(ascending=False).head()
```

Cell 57: Code

```
#Top 10 countries producing Netflix content?
df['country'].value_counts().head(10)
```

Cell 58: Code

```
#Which year had the most releases?
df['release_year'].value_counts(ascending=False).head(1)
```

Cell 59: Code

```
#Which actor appears the most?
df['cast'].value_counts(ascending=False).head(1)
```

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Business Insights

Cell 61: Code

```
country_df = df.assign(country=df['country'].fillna('Unknown').str.split(', '))
country_df = country_df.explode('country')

top_countries = country_df['country'].value_counts().head(10)

plt.figure(figsize=(12,6))
sns.barplot(x=top_countries.values, y=top_countries.index, palette='viridis')
plt.title("Top 10 Countries Producing Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.show()
```

Cell 62: Code

```
genre_df = df.assign(listed_in=df['listed_in'].str.split(', '))
genre_df = genre_df.explode('listed_in')

top_genres = genre_df['listed_in'].value_counts().head(10)

plt.figure(figsize=(12,6))
sns.barplot(x=top_genres.values, y=top_genres.index, palette='magma')
plt.title("Top 10 Genres on Netflix")
plt.show()
```

Cell 63: Code

```
movies = df[df['type']=='Movie']
```

```

movie_year = movies['release_year'].value_counts().sort_index()

plt.figure(figsize=(14,6))
sns.lineplot(x=movie_year.index, y=movie_year.values, marker='o')
plt.title("Movie Production Over Time")
plt.xlabel("Release Year")
plt.ylabel("Number of Movies")
plt.show()

```

Cell 64: Code

```

year_growth = df['release_year'].value_counts().sort_index()

plt.figure(figsize=(14,6))
sns.lineplot(x=year_growth.index, y=year_growth.values, marker='o')
plt.title("Netflix Content Growth by Release Year")
plt.show()

```

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Insights

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- Movies account for nearly 70% of Netflix content.
- The United States contributes the largest share.
- TV-MA is the most frequent rating.
- India is among the top content-producing countries.

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Conclusion

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- This project analyzed Netflix's content library using Python, Pandas, Matplotlib, and Seaborn. Through data cleaning, visualization, and business-driven analysis, key trends in content type, genre, country, ratings, and release patterns were identified. The insights can help understand Netflix's content strategy and highlight opportunities for future investment in growing markets and popular genres.

Cell 69: Code